Experiment 14: Bonding and solubility

Notes

Processing our resources often involves solvent chemistry. Bauxite for example is first mixed with sodium hydroxide. The soluble aluminium salts are dissolved, while most impurities are insoluble in the caustic solution and are separated as insoluble solids. Ethanol, a renewable resource, is fast becoming an important additive to our octane petroleum fuels. This is possible because the ethanol is soluble in hydrocarbons.

The solubility of a substance depends on the solvent being used to dissolve it. For example, sodium chloride is quite soluble in water but is insoluble in kerosene. Conversely, petrol is insoluble in water but is quite soluble in kerosene.

In this experiment water and kerosene, two very different solvents, are used. The purpose of the experiment is to compare the effectiveness of water and kerosene as solvents for solutes with different types of bonding.

Equipment

test tube rack
test tubes (nine)
beaker (100 mL)
distilled water
kerosene (20 mL)
a few crystals of the ionic substances:
sodium chloride [$NaC\ell$]
copper(II) sulfate-5-water [$CuSO_4{\cdot}5H_2O$]
calcium carbonate [$CaCO_3$]

a few crystals of the following polar covalent molecular substances:
sucrose [$C_{12}H_{22}O_{11}$]
urea [NH_2CONH_2]
ethanol [CH_3CH_2OH] (~1 mL)
a few crystals of the following non-polar covalent molecular substances:
iodine [I_2]
naphthalene [$C_{10}H_8$]
motor oil (~1 mL)

Procedure

1. Prepare a results table in which to record the solubility of the nine solutes listed above in water and kerosene. Record your results as insoluble (i), slightly soluble (ss) or soluble (s).
2. Place 1-2 mL of kerosene into each of nine test tubes.
3. Test the solubility of sodium chloride in kerosene by adding a few crystals to one of the test tubes. Shake the test tube gently and record the result in your table.
4. Repeat this procedure for each of the solutes in turn. Where the solute is a liquid add about 2-3 drops to the kerosene.
5. Discard the contents of the test tubes as directed by your teacher and clean the test tubes thoroughly.
6. Place 1-2 mL of distilled water into the cleaned test tubes and test the solubility of each of the solutes in water. Record your results.
7. Discard the contents of the test tubes as directed by your teacher and clean as before.

Processing of results and questions

1. What generalisations can you make about the solubility in water and kerosene of the solutes tested.

2. If two pieces of clothing were stained with chromium(III) chloride and lubricating grease respectively, what types of solvent would be needed to remove the stains?